

NEURON MODEL VALIDATION - held-out fault detection, Sept 2026 (#312455)

Author

This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

Pre-registered on:

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1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

Neuron models are routinely modified (refactoring, unit conversion, porting) and validated by re-running the one simulation shipped with them, a canonical regression test. We ask what fraction of real behaviour-changing faults it detects, and whether added stimulation protocols detect faults it misses. We pre-commit to publishing either outcome, including that canonical regression is adequate.

3) Describe the key dependent variable(s) specifying how they will be measured.

Detection rate per strategy: the fraction of admissible behaviour-changing faults (mutants) detected on held-out models. Detection means one of the strategy's protocols differs from the unmutated reference beyond tolerance for the same (protocol, feature) pair at step h AND at $h/2$. Tolerance $\tau = \max(\text{absolute floor, relative } x |f_h|, 3 \times |f_h - f_{h/2}|)$, calibrated on the reference so differences below the solver's own discretisation error cannot count.

4) How many and which conditions will participants be assigned to?

No human participants. Units are model faults; every strategy sees every fault (paired). Five strategies: canonical alone; a $k=4$ battery selected on development models; random count-matched; random runtime-matched; all candidate protocols. Recorded per fault: mutation family, severity (mild/strong by recorded magnitude), model.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

Estimand $\Delta = DR_{\text{selected}} - DR_{\text{canonical}}$ on held-out admissible faults. Paired counts; cluster bootstrap interval clustered on base model, $B=10,000$, seed 20260917; exact McNemar and cluster permutation ($n=10,000$) as support; two-sided. $\Delta \geq 0.10$ is the smallest material difference; below that we report no material advantage. Canonical is separately called adequate at ≥ 0.95 detection. Pooled estimates are always reported with per-model values.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

No trimming or winsorizing; the outcome is a count. Excluded from the denominator, counts reported: faults that are structurally invalid, fail to execute, are numerically unstable, or are classed equivalent within the tested domain. Tolerance entries or protocols unstable between h and $h/2$ are excluded from calibration or full-trace comparison, and recorded. Detections not reproducing at h and $h/2$ do not count. Inconclusive if control false positives exceed 0.10, over half of apparent survivors fail $h/4$ confirmation, or 25% of tolerance entries are refinement-excluded. A model whose canonical trace tolerance cannot be calibrated is reported as unclassifiable for the full-trace comparison, never counted as a survivor.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

4-6 held-out models from at least three source repositories never used in development, about 25 variants each (15-25 faults plus at least 5 behaviour-preserving controls): roughly 100-150 variants. Models qualify on thirteen pre-specified criteria applied to the unmutated model only. Size is set by compute feasibility, not power. If fewer than four qualify we stop and report the design infeasible; no criterion is relaxed.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

Valid despite existing data: no observation exists for the held-out models covered here; they have not been mutated and no detection outcome generated or inspected. Development data, declared in full: a 2-model pilot (canonical detected 22 of 22 faults, battery 19), a 5-model pilot (88 faults, 40 controls, 15 numerical stress tests), a rehearsal, and screening of unmutated behaviour only. All method parameters were fixed on that data and are frozen here; none is re-derived on held-out data.

Existing evidence is mixed and is stated now so neither outcome can later be framed as expected or surprising: canonical regression detected 22 of 22 faults on two closely related cells, but only 92% on five diverse models, missing nine at feature level, two of which survived full-trace comparison as well. Operator-validation runs had already observed the canonical outcome of nine pilot faults; pilot results are reported over all faults and over the 79 unexposed. No held-out model is affected.